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Before we start

Let’s rescue our Nginx Server

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Lesson 3

Git & Version Control

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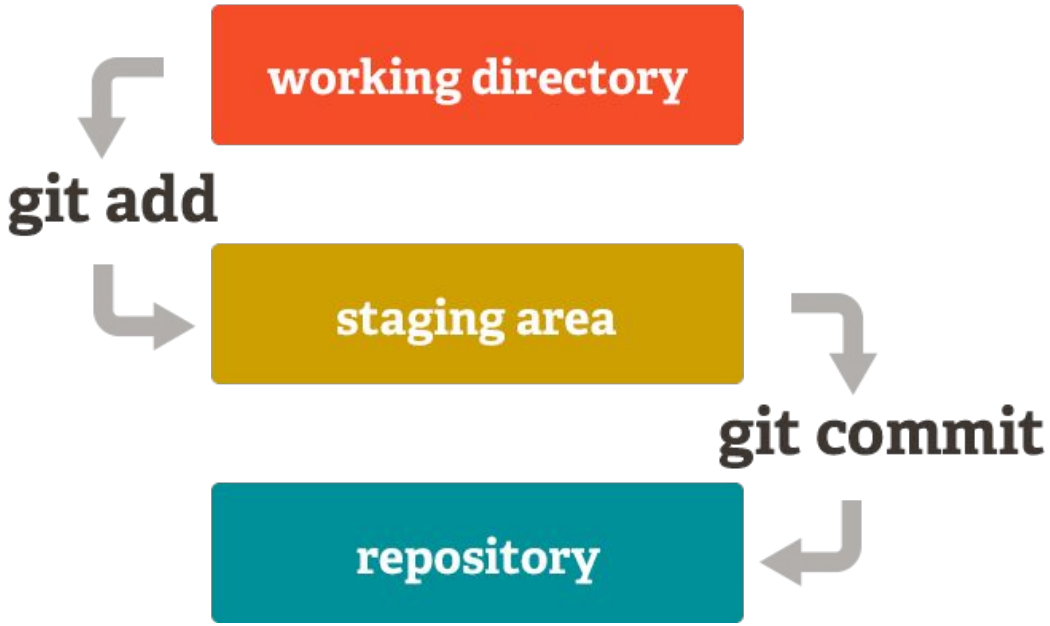
Remember him?)
Linus Torvalds: Creator of Linux & Git.

Built Git in 2005 to manage the massive Linux Kernel project because existing tools were too slow.

- **The Problem:** Naming files script_final_v2_FINAL_really.sh.
- **The Solution:** Git. Distributed, fast, and tracks every single change.

- **Working Directory:** Your actual files on your computer. It's a sandbox.
- **Staging Area:** A rough draft space. "I want these specific changes in my next snapshot."
- **Local Repository (.git):** The permanent database of your committed snapshots.



git init: Creates a hidden .git folder. Your directory is now a Git repo.
git status: Your best friend. Tells you what branch you are on and what files are modified, staged, or untracked.

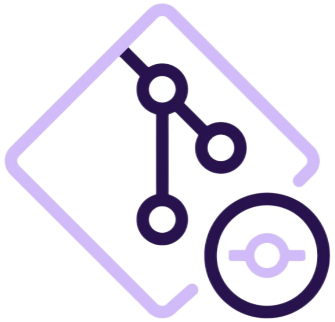
```
# The First 10 GIT COMMANDS I Type in Every New Project

git init
git status
git add .
git commit -m "Initial commit"
git branch -M main
git remote add origin https://github.com/username/repository.git
git push -u origin main
git log
```

git add <file>: Moves a file to the Staging Area. Use git add . to stage everything.

git commit -m "message": Takes a snapshot of the staging area.

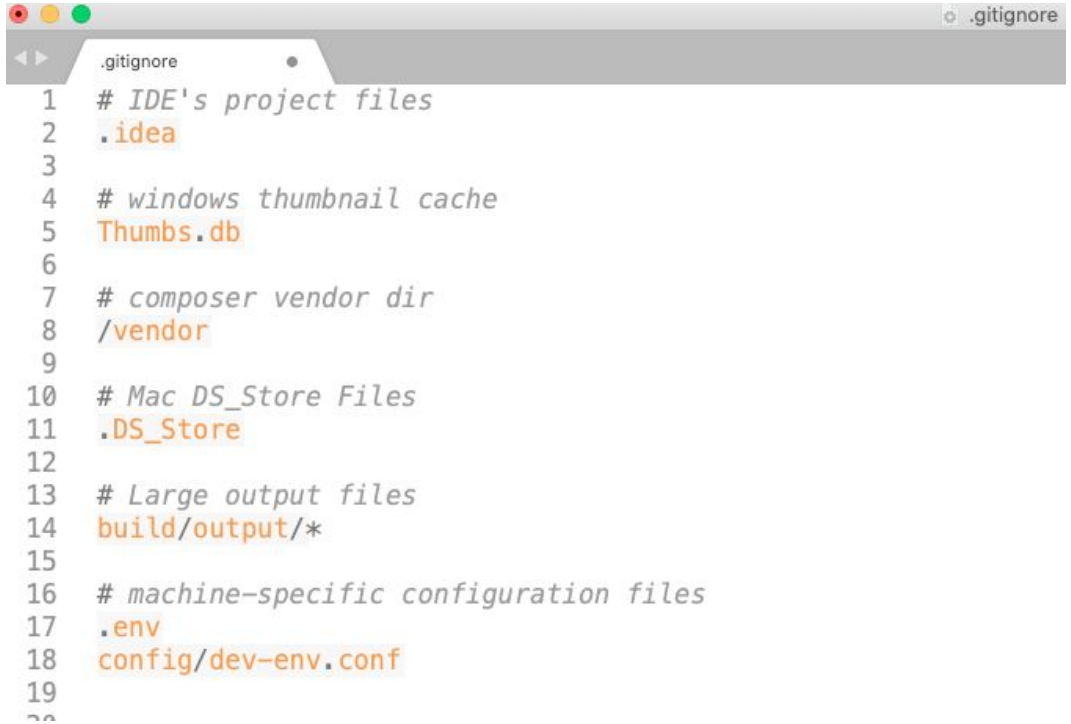
git log: Shows the history of your commits, who made them, and when.



Git Commands

.gitignore

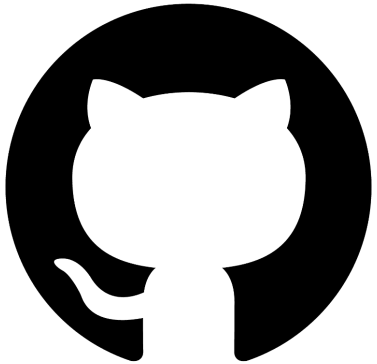
The Problem: We don't want to track logs, compiled binaries, or secret keys.
The Solution: The .gitignore file.
How it works: Any file or folder pattern listed inside .gitignore becomes invisible to Git.



Remote Repositories

Remotes: Versions of your project hosted on the internet (GitHub, GitLab, Bitbucket).
git remote add origin <URL>: Links your local repo to a remote server.
git push: Uploads your local commits to the remote repository.
git pull: Downloads changes from the remote repository to your local machine.

GitLab



GitHub

Bitbucket

Now, let’s do some real work!

Now, it’s time for the Quiz

7 Homework & Wrap-Up

- Next Lesson: Collaborative Development (Branching, Merge Conflicts, Pull Requests).
- Prep Work: Ensure your GitHub account is fully set up and you remember your credentials! We will be simulating a team environment.

Questions?

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